

## Exercise Sheet

### Task 1 – Euclidean Algorithm

Below, you will find the pseudocode for the Euclidean algorithm, which is used to compute the greatest common divisor (GCD) of two numbers greater than zero.

- a) What is the GCD of 20 and 6?
- b) Translate the pseudocode-snippet into assembly code.

```
gcd(a,b):  
  while b != 0  
    t := b  
    b := a mod b  
    a := t  
  return a
```

### Task 2 – Negabinary Numbers

The negabinary representation of a given number  $x$  is its representation to the base of -2. Given the coefficients  $a_n, a_{n-1}, \dots, a_1, a_0$  the number can be represented using this formula:

$$x = \sum_{i=0}^n a_i(-2)^i = +a_2(-2)^2 + a_1(-2)^1 + a_0(-2)^0$$

- (a) Convert  $(101101)_{-2}$  and  $(011101)_{-2}$  to their decimal representation.
- (b) Convert  $(12)_{10}$  and  $(-23)_{10}$  to their negabinary representation using 6 bits to represent the number.
- (c) List all values in negabinary and decimal representation that can be displayed with  $n = 2$ . In your own words, find a general rule to describe the range of negabinary numbers for different  $n$ .

### Task 3 – BIOS/UEFI

When looking at different types of memory you already came across BIOS and UEFI, research these two to answer the following questions:

- What are the tasks that these firmware interfaces perform?
- What are the key differences between those two?
- Where are those saved?
- Can they be updated, what difficulties might arise?